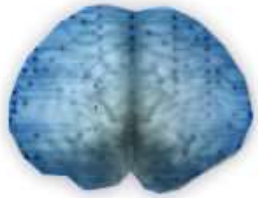


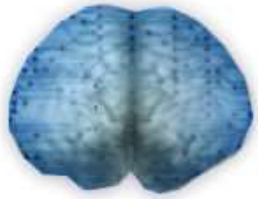
Sets and Functions

Mathematical Backgrounds of KR



Let’s make a list

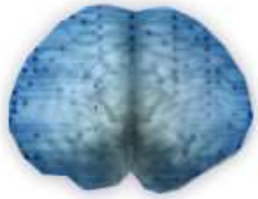
1. Pull out a piece of paper
2. Make a list of everything in this room right now
3. Now make a list of those things from step 2 that could be easily moved out
4. Now make a list of those things from step 2 that are alive



What did we do?

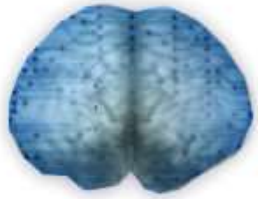
- Created a domain
- Created mappings from the domain
  - Mobile(x)
  - Alive(x)

- What ways could we implement these mappings?



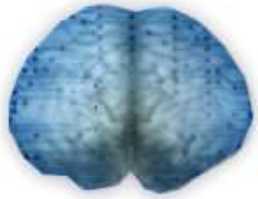
Basic Sets

- An arbitrary collection of elements
  - Numbers, points, etc.
  - Also, chairs, people, pets, geological formations, etc.



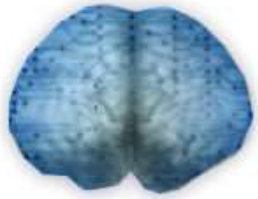
Basic Sets Notation

- Curly braces
  - e.g. {1, 97, 63, 12}
- Order is not considered
  - e.g. {12, 63, 97, 1} is equivalent to the previous



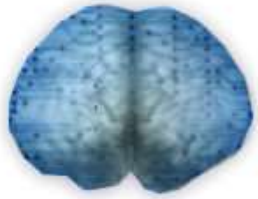
What about very large sets?

- Specification must state some rule or property
  - {x | vertebrate(x) and warmBlooded(x) and hasHair(x) and lactiferous(x)}
  - In English: “the set of all x such that x is vertebrate, x is warm blooded, x has hair, and x is lactiferous”



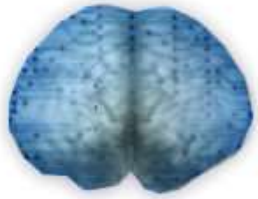
Some vocabulary

- Definition by **extension** – elements are listed explicitly
- Definition by **intension** – a specification that states a property that must be true of each element



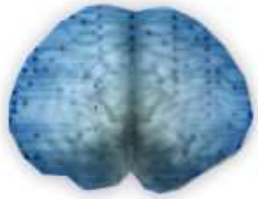
Privileged Sets

- The empty set
  - $\{\}$
- The Universal set
  - U
  - Usually the basis from which other sets are derived



The $\in$ operator

- States that a particular element is in a set
- $x \in S$ means that
  - x is an element of the set S
  - x is a member of the set S
  - x is in S



Other set operators (common)

- Union

  $A \cup B = \{x \mid x \in A \text{ or } x \in B\}$

- Intersection

  $A \cap B = \{x \mid x \in A \text{ and } x \in B\}$

- Compliment

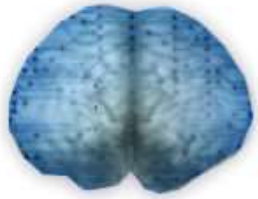
  $-A = \{x \mid x \in \boldsymbol{U} \text{ and not } x \in A\}$

- Difference

  $A\text{-}B = \{x \mid x \in A \text{ and not } x \in B\}$

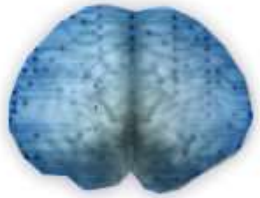
- Subset

  $A \subset B$ = If $x \in A$, then $x \in B$



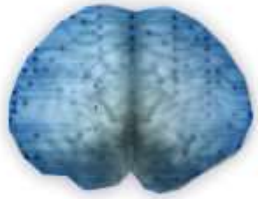
Other set operators (uncommon?)

- Proper subset
  - *A* is a proper subset of *B* if $A \subset B$ and there is at least one element of *B* that is not in *A*
- Superset
  - *A* is a superset of *B* if *B* is a subset of *A*
- Disjoint sets
  - Two sets *A* and *B* are said to be *disjoint* if they have no common elements



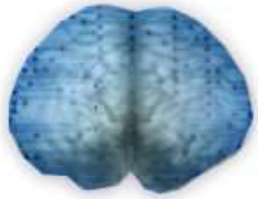
Identities

- *Idempotency*. $A \cap A$ is identical to A, and $A \cup A$ is also identical to A.
- *Commutativity*. $A \cap B$ is identical to $B \cap A$, and $A \cup B$ is identical to $B \cup A$.
- *Associativity*. $A \cap (B \cap C)$ is identical to $(A \cap B) \cap C$, and $A \cup (B \cup C)$ is identical to $(A \cup B) \cup C$.
- *Distributivity*. $A \cap (B \cup C)$ is identical to $(A \cap B) \cup (A \cap C)$, and $A \cup (B \cap C)$ is identical to $(A \cup B) \cap (A \cup C)$.
- *Absorption*. $A \cap (A \cup B)$ is identical to A, and $A \cup (A \cap B)$ is also identical to A.
- *Double complementation*. $- -A$ is identical to A.
- *De Morgan's laws*. $-(A \cap B)$ is identical to $-A \cup -B$. and $-(A \cup B)$ is identical to $-A \cap -B$.



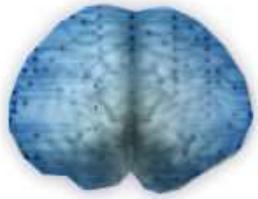
Defining complex sets

- E.g. the set of all grammatical sentences in some language
- Typically specified by a recursive definition
- Example: all positive integers not divisible by 3
  - Let the set {1, 2} be a subset of *S*.
  - If *x* is any element of *S*, then *x*+3 is also an element of *S*.
  - *S* is the smallest set that has the above two properties; i.e., *S* is a proper subset of any other set that has those properties



Bags

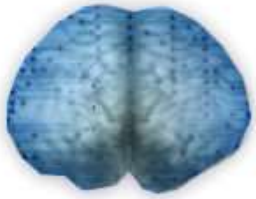
- A collection of elements that may contain duplicates
- A *sequence* is an ordered bag
  - Denoted with angle brackets
  - e.g. <178, 184, 178, 181>
- A sequence of two elements is sometimes called an *ordered pair*
- A sequence of *n* elements is an *n-tuple*



Cross product of sets

- *A*x*B*
  - Is the set of all possible ordered pairs with the first element of each pair taken from *A* and the second element from *B*
- If *A* is the set {1,2} and *B* is the set {x,y,z}

  *A*x*B* = {<1,x>,<1,y>,<1,z>,<2,x>,<2,y>,<2,z>}

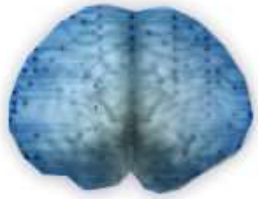


Functions

- A *function* is a rule for mapping the elements of one set to elements of another set

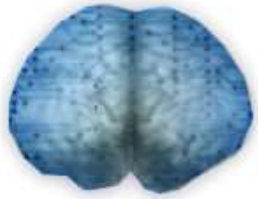
$$f: A \rightarrow B$$

- Also written as $f(x)$
- A is the domain of f
- B is the range of f
- x is the *input* or *argument*
- $f(x)$ is the *output* or *image*



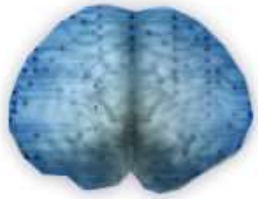
Onto functions

- A function is *onto* if every element of its range is the image of some element of its domain.
- ***Z*** is the set of all integers, and ***N*** is the set of non-negative integers
- *square*(x) is NOT onto
- *abs*(x) is onto



One-to-one functions

- A function is *one-to-one* if no two elements of its domain are mapped into the same element of its range.
- *abs*(x) is not one-to-one
  - *abs*(6) = 6 and *abs*(-6) = 6
- $g(x) = 2x2 + x$ is one-to-one
  - $g(0)=0$, $g(1)=3$, $g(-1)=1$, $g(2)=10$, $g(-2)=6$, etc.
  - $g(x)$ is not onto because many elements of ***N*** are not images of any element of ***Z***

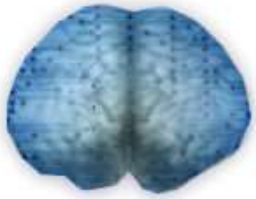


Isomorphic functions

- A function that is both one-to-one and onto
- Example:
  - Let ***E*** be the set of even integers, and let ***O*** be the set of odd integers
  - *increment*(*x*) = *x* + 1 is **isomorphic**
- If a function mapping *A* to *B* is isomorphic, then there is also an inverse of it

$$f^{-1}: B \rightarrow A$$

  - *decrement*(*x*) = *x* - 1



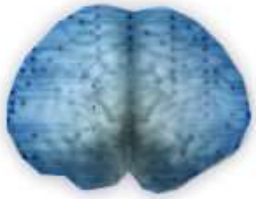
Composition of functions

- The application of one function to the result of the other

$$f: A \rightarrow B \text{ and } g: B \rightarrow C$$

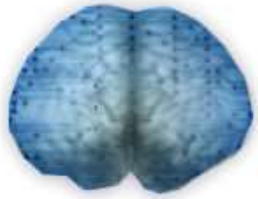
- $g(f(x))$ is a function from A to C
- A function can have more than one argument
  - The arguments can be said to be the cross product of the argument sets

$$f: A \times B \rightarrow C$$



Final vocabulary

- One-argument functions are *monadic,* two-arguments is *dyadic*, *triadic*, … *n-adic*
  - *E.g. Addition, subtraction, multiplication, and division are dyadic functions defined over the real numbers*
- “Mapping” = “function” = “operator”
- The rule that defines a mapping from two sets is called the *intention* of the function
- The set of ordered pairs that results from that rule is the *extension* of the function



Why do we care?

- Set theory is the foundation of KR
- An ontology is the definition of valid domains, ranges, and operators